Solubility Equilibria for Salts

$PbCl_2$ and AgC1 are both insoluble salts. However if chloride ion is slowly added to a solution containing both Pb^{+2} and Ag^+ ions, nearly all of the Ag^+ is precipitated as AgC1 before any Pb^{+2} separates from the solution as $PbCl_2$. This happens because AgCl is much less soluble than $PbC1_2$.

Solubiity Product Constant, K_{sp}

Let's begin with the compound AgC1. When placed in water, a small amount of this salt dissolves and the following equilibrium is established once the solution becomes saturated.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

(In a saturated solution of AgCl)

We can write an equilibrium expression following the procedure:

$$\frac{[Ag^+(aq)]\,[Cl^-(aq)]}{[AgCl(s)]} = K_c$$

[Ag(aq)] [C1⁻(aq)] = K_c [AgCl(s)] = a new constant

This new constant is called the solubility product constant, because it equals the product of concentration terms for the ions dissolved in a saturated solution of a sparingly soluble substance. There is no denominator left in the mass action expression, and the equilibrium condition in a saturated solution of AgCl is:

$$[Ag^+]\,[C1^-] = K_{sp}$$

The mass action expression in this kind of solubility equilibrium is often called the ion product because it is a product of ion concentrations.

The solubilities of salts change with temperature.

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Many salts produce more than one of a given ion when they dissociate Silver chromate, Ag_2CrO_4, is an example. Each of its formula units gives two Ag^+ ions and one CrO_4^{-2} ion. Its solubility equilibrium is:

$$Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)$$

The solubility product expression for Ag_2CrO_4 is:

$$[Ag^+]^2[CrO_4^{2-}] = K_{sp}$$

Calculating K_{sp} from Molar Solubility Data

EX.4

At 25°C, one L of water can dissolve 7.1×10^{-7} mol of AgBr. Calculate the K_{sp} of AgBr at this temperature.

Analysis

$$AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq)$$

$$K_{sp} = [Ag^+][Br^-]$$

We will construct a concentration table

	$AgBr(s)$	$\rightleftharpoons$	$Ag^+(aq)$	+	$Br^-(aq)$
Initial concentrations (M)	No entries in this column		0 (Note 1)		0
Changes in concentrations when AgBr dissolves (M)			$+7.1 \times 10^{-7}$ (Note 2)		$+7.1 \times 10^{-7}$
Equilibrium concentrations (M)			7.1×10^{-7} (Note 3)		7.1×10^{-7}

$$K_{sp} = [Ag^+][Br^-]$$
$$= (7.1 \times 10^{-7})(7.1 \times 10^{-7})$$
$$= 5.0 \times 10^{-13}$$

The K_{sp} of AgBr is thus 5.0×10^{-13} at 25 °C.

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EX.5

The molar solubility of silver chromate, Ag_2CrO_4 ,in pure water is 6.7×10^{-5} mol L^{-1} at 25 °C. What is K_{sp} for Ag_2CrO_4? Calculating from Molar

Analysis

The equilibrium equation and K_{sp} expression are:

$$Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq) \qquad K_{sp} = [Ag^+]^2[CrO_4^{2-}]$$

When one liter of pure water dissolves 6.7×10^{-5} mol of Ag_2CrO_4, we can see by the stoichiometry of the equilibrium equation that we must obtain 6.7×10^{-5} mol of CrO_4^{-2} and $2 \times (6.7 \times 10^{-5}$ mol) of Ag^+.

Solution

With this information, we prepare the concentration table.

	$Ag_2CrO_4(s) \rightleftharpoons$	$2Ag^+(aq)$	+	$CrO_4^{2-}(aq)$
Initial concentrations (M)		0		0
Changes in concentrations when Ag_2CrO_4 dissolves (M)		$+2 \times (6.7 \times 10^{-5}) = +1.3 \times 10^{-4}$		$+6.7 \times 10^{-5}$
Equilibrium concentrations (M)		1.3×10^{-4}		6.7×10^{-5}

Substituting the equilibrium concentrations into the mass action expression for K_{sp} gives

$$K_{sp} = (1.3 \times 10^{-4})^2(6.7 \times 10^{-5})$$
$$= 1.1 \times 10^{-12}$$

So the of Ag_2CrO_4 is 1.1×10^{-12} at 25°C.

EX.6

At 25°C, the molar solubility of $PbCl_2$ in a 0.1M NaC1 solution is 1.7×10^{-3} mol L^{-1}. Calculate the for $PbC1_2$.

Analysis

Again, we begin by writing the equation for the equilbrium and the K_{sp} expression.

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$$PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^-(aq) \qquad K_{sp} = [Pb^{2+}][Cl^-]^2$$

When the $PbCl_2$ dissolves in the NaC1 solution, the Pb^{+2} concentration increases by 1.7×10^{-3}M and the C1 concentration increases by $2 \times (1.7 \times 10^{-3}$M). With these data build the concentration table.

$$PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) \quad + \quad 2Cl^-(aq)$$

Initial concentrations (M)	0	0.10
Changes in concentrations when $PbCl_2$ dissolves (M)	$+1.7 \times 10^{-5}$	$+2 \times (1.7 \times 10^{-5}) = +3.4 \times 10^{-5}$
Equilibrium concentrations (M)	1.7×10^{-5}	0.10 + 0.0034 = 0.10 (rounded to the second decimal place)

Solution

Substituting the equilibrium concentrations into the K_{sp} expression gives

$$K_{sp} = (1.7 \times 10^{-5})(0.10)^2$$
$$= 1.7 \times 10^{-5}$$

The of $PbC1_2$ is 1.7×10^{-5} at 25 °C.

EX.7

What is the molar solubility of AgC1 in pure water at 25 °C?

Analysis

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq) \qquad K_{sp} = [Ag^+][Cl^-] = 1.8 \times 10^{-10}$$

The value of K_{sp} is from Table. Now we can build a concentration table.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

Initial concentrations (M)	0	
Changes in concentration when AgCl dissolves (M)	$+x$	$+x$
Equilibrium concentrations (M)	x	x

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Solution

We make the substitutions into the K_{sp} expression:

$$(x)(x) = 1.8 \times 10^{-10}$$

$$x = 1.3 \times 10^{-5}$$

Thus the calculated molar solubility of AgCl in water at 25°C is:

1.3×10^{-5} mol L^{-1}

EX.8

Calculate the molar solubility of lead iodide, PbI_2, in water at 25 °C.

Analysis

The relevant equations and data (from Table) are

$$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2I^-(aq) \qquad K_{sp} = [Pb^{2+}][I^-]^2 = 7.9 \times 10^{-9}$$

	$PbI_2(s) \rightleftharpoons$	$Pb^{2+}(aq)$	$+ 2I^-(aq)$
Initial concentrations (M)		0	0
Changes in concentrations when PbI_2 dissolves (M)		$+x$	$+2x$
Equilibrium concentrations (M)		x	$2x$

Solution

Substituting,

$$K_{sp} = (x)(2x)^2 = 4x^3 = 7.9 \times 10^{-9}$$

$$x^3 = 2.0 \times 10^{-9}$$

$$x = 1.3 \times 10^{-3}$$

Thus, the molar solubility of PbI_2 is 1.3×10^{-3} mol L^{-1}.

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The Common Ion Effect and Solubility

Suppose that we stir some calcium carbonate with water long enough to establish the following equilibrium.

$$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$$

Then we add to the solution a very soluble salt of calcium, like $CaCl_2$. This puts additional Ca^{+2} into solution, and it upsets the above equilibrium. The ion product is no longer equal to K_{sp} .

The common ion effect can dramatically lower the solubility of a salt, as shown in the next example.

EX.9

What is the molar solubility of PbI_2 in a 0.1 M NaI solution?

Analysis

As usual, we begin with the chemical equation for the equilibrium and the appropriate K_{sp} expression. The value of K_{sp} is obtained

$$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2I^-(aq) \qquad K_{sp} = [Pb^{2+}][I^-]^2 = 7.9 \times 10^{-9}$$

Here is the concentration table.

	$PbI_2(s) \rightleftharpoons$	$Pb^{2+}(aq)$	$+ 2I^-(aq)$
Initial concentrations (M)		0	0.10
Changes in concentrations when $PbCl_2$ dissolves (M)		$+x$	$+2x$
Equilibrium concentrations (M)		x	$0.10 + 2x$

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Solution

Substituting equilibrium values into the K_{sp} expression gives

$$K_{sp} = (x)(0.10 + 2x)^2 = 7.9 \times 10^{-9}$$

2x will be much smaller than 0.10. If this is so, then

$$0.1 + 2x \approx 0.1$$

Substituting 0.1 M for the I^- concentration gives

$$K_{sp} = (x)(0.10)^2 = 7.9 \times 10^{-9}$$

$$x = \frac{7.9 \times 10^{-9}}{(0.10)^2}$$

$$= 7.9 \times 10^{-7}\ M$$

Thus, the molar solubility of PbI_2 in 0.10 M NaI is 7.9 x10^{-7}M

Determining Whether a Precipitate Will Form in a Solution

When comparing a computed ion product with the for the salt, we can tell whether a particular combination of ion concentrations will produce a precipitate or not. Here are the possibilities.

Precipitate will form	Ion product > K_{sp} (supersaturated)
No precipitate will form	Ion product = K_{sp} (saturated) Ion product < K_{sp} (unsaturated)

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EX.10

A student wished to prepare 1 .0 L of a solution containing 0.015 mol of NaCl and 0.15 mol of $Pb(NO_3)_2$. Knowing from the solubility rules that the chloride of Pb^{+2} is "insoluble," there was concern that a precipitate of $PbCl_2$ might form. Will it?

Analysis

By writing the solubility equilibrium and the K_{sp} expression that applies to a saturated solution.

$$PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^-(aq) \qquad K_{sp} = [Pb^{2+}][Cl^-]^2$$

From Table, $K_{sp}=1.7 \times 10^{-5}$.

Solution The planned solution would have the following concentrations:

$$[Pb^{+2}] = 0.15 \text{ mol } L^{-1} = 0.15 \text{ M}$$

$$[Cl^-] = 0.015 \text{ mol } L^{-1} = 0.015 \text{ M}$$

We use these values to compute the ion product for $PbCl_2$.

$$[Pb^{+2}][Cl^-]^2 = (0.15)(0.015)^2 = 3.4 \times 10^{-5}$$

Notice that this value is larger than the K_{sp} of $PbCl_2$, 1.7×10^{-5}. This means a precipitate of $PbCl_2$ will form if the preparation of this solution is attempted.

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